

Multiple Choice (10 marks)

1. Of the cans of soup in Rolando's cupboard, $\frac{1}{2}$ are tomato and $\frac{2}{5}$ are chicken noodle. What fraction of the cans of soup in Rolando's cupboard are either tomato or chicken noodle?
 - (a) $\frac{2}{3}$
 - (b) 10-Sep
 - (c) 5-Mar
 - (d) $\frac{6}{11}$
 - (e) $\frac{5}{8}$
2. The value of the digit 5 in 24,513 is how many times the value of the digit 5 in 357?
 - (a) 0
 - (b) 50
 - (c) 100
 - (d) 100,000
 - (e) 5,000
3. If a series circuit has a capacitor of $C = 0.8 \times 10^{-6}$ F and an inductor of $L = 0.2$ H, find the resistance R so that the circuit is critically damped.
 - (a) 500Ω
 - (b) 400Ω
 - (c) 3000Ω
 - (d) 1250Ω
 - (e) 1000Ω
4. Let V be the space spanned by functions $\cos(2x)$ and $\sin(2x)$. Find the determinant of the linear transformation $D(f) = f'$ from V to V .
 - (a) 0
 - (b) -1
 - (c) 1
 - (d) 2
 - (e) -2
5. How many ways are there to partition a set of 5 elements into 3 non-empty cycles?

- (a) 45
- (b) 10
- (c) 15
- (d) 40
- (e) 50

True / False (10 marks)

Answer True or False

6. True or False: The correct answer to '5.4-21. Let X and Y be independent with distributions $N(5, 16)$ and $N(6, 9)$, respectively. Evaluate $P(X > Y) = P(X - Y > 0)$.' is '0.5873'.
7. True or False: The number 343 can be expressed in exponential notation as 7^3 .
8. True or False: Reflecting the vertex $(-1, 4)$ across the line $y = x$ results in the vertex $(4, -1)$ in pentagon P' .
9. True or False: The correct answer to ' $\lim_{x \rightarrow 1} \{1/(x - 1) - c/(x^3 - 1)\}$ exists. What is the value of c ?' is '4'.
10. True or False: The largest integer for which $(x+11)/(x+7)$ is an integer is 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of a correlation coefficient of 1 in the context of statistical analysis, particularly focusing on its interpretation regarding cause-and-effect relationships between variables.
12. Discuss the implications of a 70% probability of rain on a given day. What is the complementary probability of no rain, and how does this concept apply to probability theory?

End of Paper